



Numerical Simulation and Field Experiment of Rock-Breaking of PDC-Cone Hybrid Bit

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Abstract: In order to study polycrystalline diamond compact (PDC)-cone hybrid bit penetrating rock formation, a series of mechanism analysis, numerical simulations and field drilling experiment are conducted in this paper. Firstly, the rock-breaking mechanism of hybrid bit is analyzed, including the mutual contribution of the roller-cone and PDC cutting structures to rock destruction. On this basis, the 3D finite element (FE) modeling of hybrid bit and PDC bit penetration rock formation is performed. In this process, the 3D uniaxial compression experiment and simulation are carried out to calibrate and determine the key property parameters of granite. Then, the torsional and axial dynamics characteristics of hybrid bit and PDC bit are quantified for comparison. The results show that, under the same drilling conditions, the torque fluctuation of the hybrid bit is reduced by up to 50% compared with that of the PDC bit. The hybrid bit also helps to mitigate torsional vibration and suppress stick-slip behavior. It can also increase the axial vibration amplitude without essentially leading to the rapid failure of drilling tool. Therefore, hybrid bit has higher rock-breaking efficiency in hard formations, which is in good agreement with the field drilling experiment results.

Keywords: Hybrid bit; Rock-breaking; Mechanism analysis; Numerical simulation; Field experiment

1 Introduction

Drill-bit is one of the core tools of oil and gas drilling, and development of drill bit technology will directly improve drilling efficiency and reduce drilling costs. In most drilling operations today, polycrystalline diamond compact (PDC) bit and roller-cone bit are the most commonly used. PDC bit, in particular, has been used for more than 90% global drilling length due to its great advantages in terms of drilling speed and wear resistance [1]. However, in the context of the current shift in the main battlefield of oil and gas exploration and development to deep and ultra-deep formations, the hardness and plasticity of rock increase with the increase of well depth, while both the rate of penetration (ROP) and service life of PDC and roller-cone bits decrease, causing low rock-breaking efficiency in hard formations, as shown in Figure 1. Hence, it is urgent to develop a new type of drill-bit which can crush hard formations more effectively.

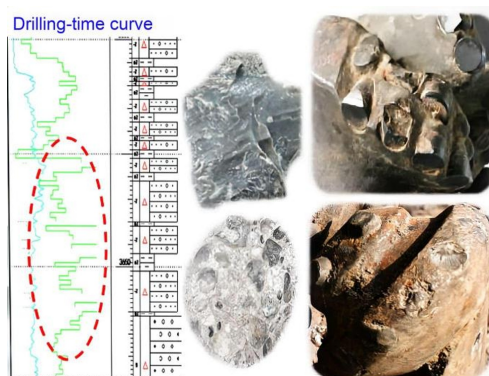


Figure 1. Low efficiency and serious failure of polycrystalline diamond compact (PDC) and roller-cone bits

Hybrid drill-bit technology combines PDC and roller-cone bits into an integrated unit to make the most of the advantages of each drill-bit type, has led to a completely new idea to improve ROP in hard formations and reduce drilling costs. Specifically, the roller-cone is set up as a secondary cutting structure to provide a cushion to protect the PDC cutters, thereby increasing drilling speed [2, 3]. The concept of the PDC-cone hybrid bit dates back to the 1930s, but it wasn't until the PDC cutter technology advanced considerably that manufacture of a reliable hybrid bit had become feasible. In 2009, Baker Hughes successfully launched a new generation of PDC-cone hybrid bit, which have since seen numerous applications in drilling operations [4–7]. The results showed that PDC-cone hybrid bit, combined with the automated drilling system, provided better effects in terms of ROP, verticality control, distance drilling and stability than PDC and roller-cone bits. On the other hand, the design methods [8, 9], working load characteristics [10, 11] and torsional vibration reduction mechanism [12, 13] of PDC-cone hybrid bit have also attracted the attentions of various scholars.

The remaining sections of this paper are organized as follows. Section 2 discusses the rock-breaking mechanism of PDC-cone hybrid bit, including the effects of cushion and pre-cracking of roller-cone structure. Then the finite element (FE) models used to simulate PDC-cone hybrid bit and PDC bit breaking rock are developed in Section 3, and the results and discussion of numerical simulation are given in Section 4. Subsequently, a field experiment of PDC-cone hybrid bit is performed in Section 5. Finally, some conclusions are drawn in Section 6.

2 Rock-Breaking Mechanism of Hybrid Bit

The major structures of a PDC-cone hybrid bit include drill-bit body, PDC cutting structure, roller-cone structure, bearing system and hydraulic system, as shown in Figure 2. The central portion of the borehole is cut solely by the PDC cutting structure while the outer portion which is more difficult to drill is disintegrated by the combined action of the roller-cone structure and PDC cutting structure.



Figure 2. Major structures of polycrystalline diamond compact (PDC)-cone hybrid bit

When PDC cutting structure breaks rock, the PDC cutter penetrates the formation under the action of weight on bit (WOB) and shears rock persistently under the driving effect of rotary table, causing elastic strain in the contact areas between rock and the PDC bit cutter. Along with its constant movement, the powerful stress is big enough to cause plastic strain when the shear stress exceeds the shear strength of rock. Then large shear and squeeze deformation of rock will appear caused by the frictional force and pressure from the rake face of PDC cutter, as shown in Figure 3a. In an ideal situation, all PDC cutters scrape rock in a spiral movement along the axes of drill-bit. Therefore, the bottom hole morphology caused by PDC cutting structure shows as continuous concentric trails, as shown in Figure 3b.

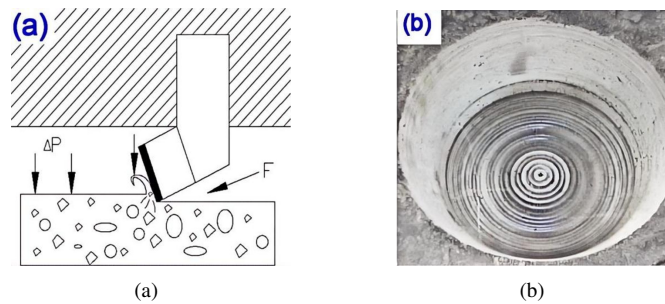


Figure 3. Rock fragmentation caused by polycrystalline diamond compact (PDC) cutting structure: (a) schematic diagram; (b) bottom hole morphology

During the process of roller-cone structure breaking rock, rock is crushed by the roller-cone cutter under the action of WOB. Crushed rock will generate lateral pressure to surrounding rock due to the presence of WOB, which is deemed as the cause of the shear stress. The internal cracks of rock will extend to rock surface along the direction of the maximum shear stress and craters will be formed when the shear stress exceeds the shear strength of rock, as shown in Figure 4a. To sum up, the bottom hole morphology caused by the roller-cone structure manifests as discrete breaking pits, as shown in Figure 4b.

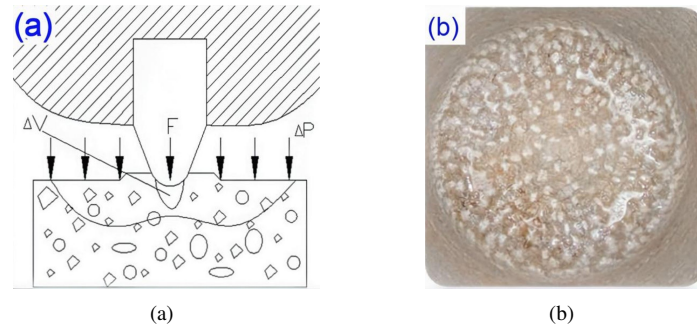


Figure 4. Rock fragmentation caused by roller-cone structure: (a) schematic diagram; (b) bottom hole morphology

Figure 5a represents the schematic diagram of PDC-cone hybrid bit breaking rock. Due to the impact crushing effect of roller-cone cutter on rock, craters are formed and the bottom-hole rock gets rough in this region, which really benefits subsequent PDC cutter. Furthermore, the impact crushing effect of roller-cone cutter brings about not only craters but also shock cracks, as shown in Figure 5b. The foregoing rock damage can reduce rock strength, which increases rock cutting efficiency and prolongs service life of PDC cutter. Due to the combined effect of both PDC cutting structure and roller-cone structure, the bottom hole morphology caused by hybrid bit integrates the characteristics of that respectively caused by these two types of cutting structures, as shown in Figure 5c.

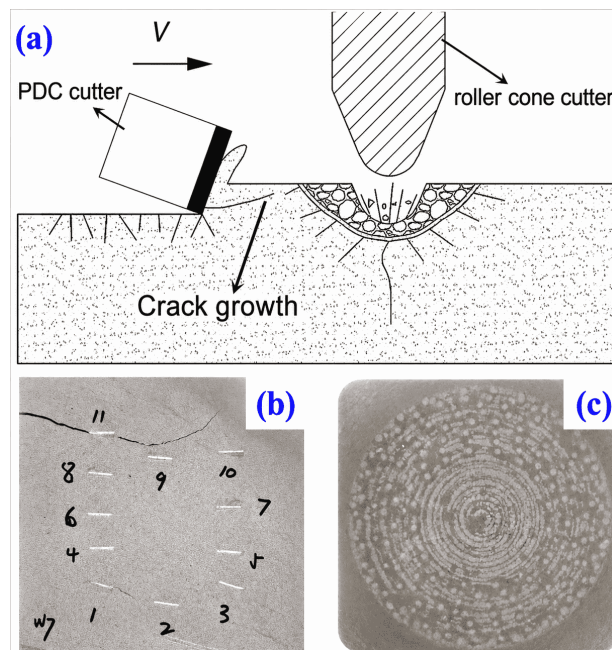


Figure 5. Rock fragmentation caused by polycrystalline diamond compact (PDC)-cone hybrid bit: (a) schematic diagram; (b) shock cracks caused by roller-cone cutter; and (c) bottom hole morphology

When a PDC-cone hybrid bit penetrates a formation, the rock prebreaking caused by the roller-cone structure can reduce the energy for the PDC cutting structure scraping rock. This can reduce the operation torque and its fluctuation obviously, improving directional performance. Meanwhile, the roller-cone structure will inevitably share a portion of WOB and limit the penetration depth of PDC cutters, so as to protect the PDC cutting structure and improve its impact resistance.

3 Numerical Simulation of Rock-Breaking

In this section, the numerical simulations of hybrid bit and PDC bit penetrating rock formation under real drilling conditions are implemented on the basis of finite element method (FEM). For improving calculation efficiency, four basic assumptions are made: (1) Rock is assumed to be an isotropic material; (2) Only the cleaning and cooling effects of drilling fluid on rock and drill-bits are taken into consideration; (3) The elements are instantly removed from the rock matrix after failure; (4) Drill-bits have much higher strength and hardness than rock, so both PDC-cone hybrid bit and PDC bit are assumed to be rigids.

3.1 Constitutive Relation and Petrophysical Parameters of Lithology

The Drucker-Prager strength criterion is introduced to characterize the elastic-plastic behavior of the rock used in the later simulations. The Drucker-Prager strength criterion, which can reflect the impact of volume stress on rock strength, treats deviatoric stress as the primary cause of rock damage. It has considered the effect of hydrostatic pressure on yield process and has been widely applied in research for the process of cutting rock. According to the Drucker-Prager strength criterion, the intermediate principal stress has an influence on rock damage, which is expressed by the normal stress σ_{oct} and the shear stress τ_{oct} in a regular octahedron [14]. The shear failure model can be expressed by using the principal stresses as follows [15]:

$$\tau_{oct} = \tau_0 + m\sigma_{oct} \quad (1)$$

in which,

$$\begin{aligned} \tau_{oct} &= \frac{1}{3} \sqrt{(\sigma_1 - \sigma_2)^2 + (\sigma_2 - \sigma_3)^2 + (\sigma_3 - \sigma_1)^2} \\ \sigma_{oct} &= \frac{1}{3} (\sigma_1 + \sigma_2 + \sigma_3), \\ m &= -\sqrt{6}\alpha, \tau_0 = \frac{\sqrt{6}}{3}k \end{aligned} \quad (2)$$

where, σ_1, σ_2 and σ_3 are respectively the maximum principal stress, intermediate principal stress and minimum principal stress of rock element under the external load. k is a material constant related to grain cementation and α is another material constant related to the friction between different grains α and k can be fitted by the cohesion c and the internal friction angle φ of rock. When the stress lode angle θ_σ takes different values, the fitting results are not the same.

When $\theta_\sigma = \frac{\pi}{6}$, as the compression hardening, α and k can be expressed as follow:

$$\alpha = \frac{2 \sin \varphi}{\sqrt{3}(3 - \sin \varphi)}, k = \frac{6c \cos \varphi}{\sqrt{3}(3 - \sin \varphi)} \quad (3)$$

When $\theta_\sigma = -\frac{\pi}{6}$, as the tensile hardening, α and k can be expressed as follow:

$$\alpha = \frac{2 \sin \varphi}{\sqrt{3}(3 + \sin \varphi)}, k = \frac{6c \cos \varphi}{\sqrt{3}(3 + \sin \varphi)} \quad (4)$$

When $\tan \theta_\sigma = -\frac{\sin \varphi}{\sqrt{3}}$, as the shear hardening, α and k can be expressed as follow:

$$\alpha = \frac{\sin \varphi}{\sqrt{3}(3 + \sin^2 \varphi)}, k = \frac{\sqrt{3}c \cos \varphi}{\sqrt{3 + \sin^2 \varphi}} \quad (5)$$

The lithology used in the simulations of this paper is granite. In order to calibrate and determine its property parameters, the 3D uniaxial compression experiment is firstly performed on the apparatus shown in Figure 6a, and then the 3D uniaxial compression simulation is carried out according to the experiment. The FE model consists of a rock cylinder in the size of $\phi 25.43 \text{ mm} \times 50.12 \text{ mm}$ and two rigid indenters in the size of $\phi 50 \text{ mm} \times 3 \text{ mm}$, as shown in Figure 6b. In particular, it should be noted that the down indenter 1 is fully constrained and a vertical downward displacement load of 0.3 mm is applied to the upper indenter. Finally, the granite sample undergoes a typical shear damage (shown in Figure 6c), which is generally consistent with the experimental phenomenon.

The stress-strain curve of granite obtained from numerical simulation has a good match with that obtained from experiment, as shown in Figure 6d. This allowed us to determine the key mechanical property parameters of granite used in the subsequent simulations, as shown in Table 1.

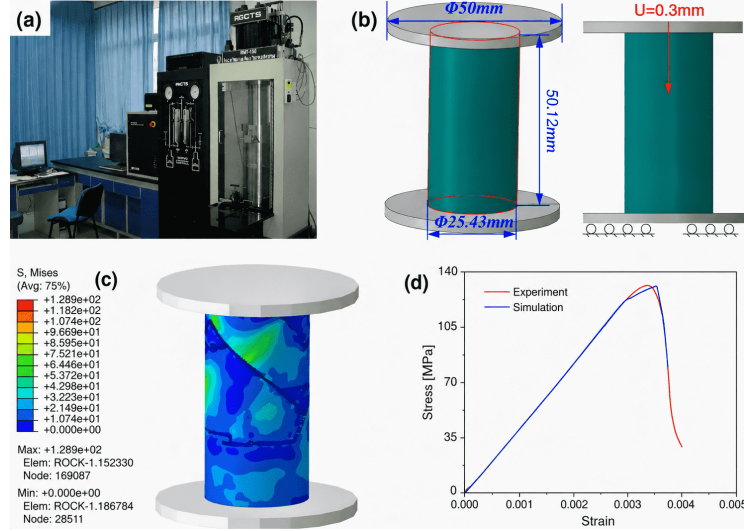


Figure 6. Uniaxial compression test and simulation of granite: (a) triaxial testing apparatus; (b) 3D finite element (FE) model; (c) damage contour; and (d) stress-strain curves from experiment and simulation

Table 1. The calibrated mechanical property parameters of granite

Material	Parameter	Value	Parameter	Value
Granite	Density [t/mm^3]	2.84×10^{-9}	Angle of dilatancy [$^\circ$]	10
	Elasticity modulus [MPa]	43080	Yield stress [MPa]	121.5
	Poisson's ratio [/]	0.192	Fracture strain [/]	4.15×10^{-4}
	Angle of friction [$^\circ$]	42.95	Fracture energy [N/mm]	0.1226

3.2 The 3D Finite Element Modeling

The FE models are developed on the commercial software Abaqus. Both the hybrid bit and PDC bit have an external diameter of ϕ 216 mm, and the rock is a cylinder of ϕ 500 mm $\times$ 250 mm. For the hybrid bit, a series of hinged connections are defined to guarantee the relative motion relationship between the roller-cone structures and the drill-bit body. The two types of drill bits are discretized using 4-node linear tetrahedron elements (C3D4), while the rock is meshed with 8-node hexahedral reduced-integration elements (C3D8R). To simulate the actual drilling conditions as accurately as possible, a constant WOB and a constant rotary speed are applied to both types of drill bits. Translational and rotational displacements of the drill bits in the horizontal directions (X and Y) are fully constrained, while the constant WOB and rotary speed are imposed along the drilling direction (Z), as shown in Figure 7. It should be noted that the torque is not prescribed as a constant input; instead, it is obtained as the reaction moment at the bit-rock interface and is recorded throughout the simulation, allowing its fluctuation to be analyzed in Section 4.1.

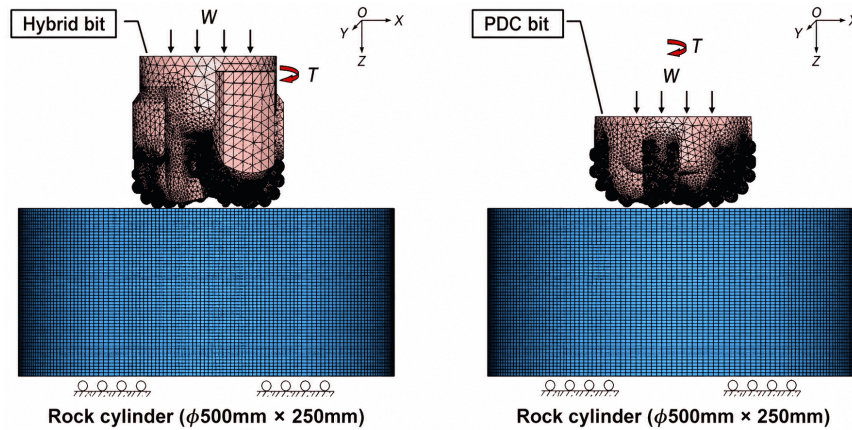


Figure 7. The 3D finite element (FE) model configurations of hybrid bit and polycrystalline diamond compact (PDC) bit penetrating rock formation

3.3 Dynamical Model of Drill-bit/rock Interaction

Drill-bit penetrating rock formation is an extremely complex process that involves dynamical contact with accompanying random collisions. The whole system is subject to the geometric nonlinearity caused by both large displacements and rotations of the drill-bit in a short period of time, as well as the material nonlinearity caused by large strains in the rock material up to failure. Meanwhile, the contact nonlinearity caused by two scenarios above is also present in this process. Using the FEM, the space domain Ω occupied by the whole contact system at moment t is established. By introducing the body force ($\mathbf{b}$), boundary force ($\mathbf{r}$ and $\mathbf{r}_c$), and Cauchy stress ($\boldsymbol{\sigma}$) acting in the contact system, the contact problem can be summarized as:

$$\int_{\Omega} \sigma \delta e d\Omega - \int_{\Omega} b \delta u d\Omega - \int_{\Gamma_f} r \delta e dS - \int_{\Gamma_c} r_c \delta u dS + \int_{\Omega} \rho a \delta u d\Omega = 0 \quad (6)$$

where, Γ_f is the boundary for a given boundary force. Γ_c is the contact boundary δu is the virtual displacement. δe is the virtual strain. a is the acceleration. Discretizing the space domain Ω by the FEM and then introducing the virtual displacement field, we can obtain the following:

$$\mathbf{M}\ddot{\mathbf{u}} = \mathbf{p}(t) + \mathbf{c}(u, \theta) - \mathbf{f}(u, \beta) \quad (7)$$

where, $\mathbf{M}$ is the mass matrix, $\ddot{\mathbf{u}}$ is the acceleration vector, $\mathbf{p}$ is the external force vector, $\mathbf{c}$ is the contact and friction force vector, $\mathbf{f}$ is the internal stress vector, u is the body displacement vector, θ is a variable associated with the contact surface characteristics, and β is a variable associated with the constitutive relation of the material.

4 Results and Discussion

4.1 Drill-bit Torsional Dynamics Characteristics

The torque comparison of hybrid bit and PDC bit in simulation process is shown in Figure 8a. These two torque curves consist of two stages, which are named the rapidly drilling stage and the stably drilling stage, respectively. At the beginning, the resistance of rock to hybrid bit and PDC bit is small because the rock surface is relatively smooth. As the drill-bits move, their contact area with the rock enlarges, and its resistance to drill-bits increases. The torque curves reach the stable stage after drill-bits are completely submerged. In this stage ($t = 2-5$ s), torque fluctuation always exists in the rock-breaking progresses both of hybrid bit and of PDC bit due to the complex drilling conditions. In the simulation, the torque fluctuation of the hybrid bit is approximately 50% lower than that of the PDC bit. This reduction trend is qualitatively consistent with the field observation that the hybrid bit achieved higher ROP and smoother drilling performance than the offset well, although direct comparison of downhole torque data is not available.

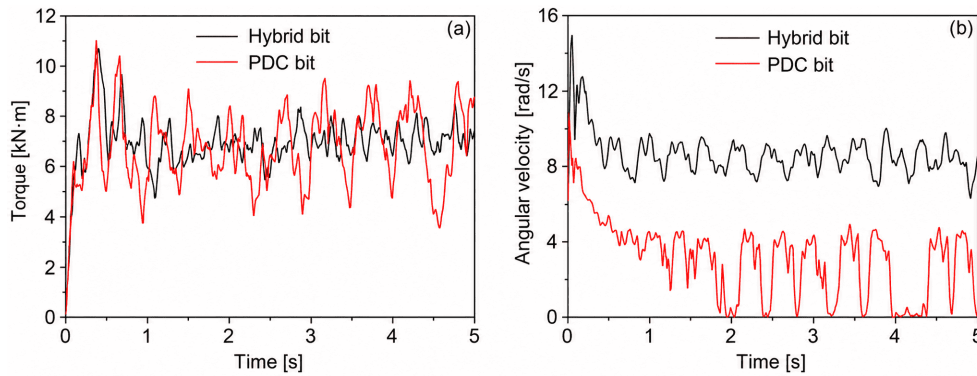


Figure 8. Torsional response comparisons of hybrid bit and polycrystalline diamond compact (PDC) bit: (a) torque; (b) angular velocity

Figure 8b displays the angular velocity curves of hybrid bit and PDC bit, which can capture the key properties of torsional vibration. The angular velocities of both hybrid bit and PDC bit fluctuate continuously, but the fluctuation range of hybrid bit is far below that of PDC bit. It should be noted that the PDC bit even undergoes typical stick-slip vibration in the stick phase, the drill-bit is forced to stop rotation when the energy is not enough to break rock, and the drill-string gathers energy constantly due to the driving of rotary table. Once the energy reaches the critical value, the drill-bit is released and immediately changes back to the slip phase, in which the drill-bit oscillates back and forth around its axis line at high angular velocity [16, 17]. Stick-slip vibration can not only cause the fatigue fracture of drill-string, but can also greatly increase the wear and fracture of drill-bit cutting teeth [18], as shown in Figure 9. In contrast, there is no significant stick-slip phenomenon in the drilling process of the hybrid bit.



Figure 9. Typical failure cases of drilling tools: (a) the fatigue fracture of drill-pipe; (b) the wear and fracture of drill-bit cutting teeth

In order to assess the intensity of drill-bit torsional vibration quantitatively, a dimensionless index S_e is introduced, and its expression is given by:

$$S_e = \frac{R_{max} - R_{min}}{2R_{top}} \quad (8)$$

where, R_{max} and R_{min} are the maximum and minimum rotational velocities of drill-bit, while R_{top} represents its average angular velocity. Meanwhile, it is specified that larger value of S_e corresponds to a more intense torsional vibration.

Therefore, the values of S_e , corresponding to the simulation results of hybrid bit and PDC bit shown in Figure 8b are 0.221 and 1.032, respectively. This indicates that employing PDC-cone hybrid bit helps to mitigate torsional vibration and consequently mitigate stick-slip phenomenon. According to the rock-breaking mechanism analysis given in Section 2, the roller-cone structures of hybrid crush rock at first, reducing the strength of rock, and then the PDC cutting structure scrapes rock efficiently and continuously. So, on the other hand, employing PDC-cone hybrid bit contributes to the protection of drilling tools and thus to their longevity.

4.2 Drill-bit Axial Dynamics Characteristics

Figure 10a shows the drilling depth curves of hybrid bit and PDC bit. In this figure, the drilling depth of hybrid bit is bigger than that of PDC bit under the same WOB and rotary speed. This illustrates that hybrid bit has higher rock breaking efficiency than PDC bit under the same drilling conditions, which is consistent with the conclusions obtain in Section 2. Furthermore, the process of drill-bit penetrating rock is not continuous, and the borehole is deepened intermittently, which is consistent with the phenomenon captured in the previous simulation. However, hybrid bit has more continuous drilling process than PDC bit. Figure 10b shows the axial acceleration curves of hybrid bit and PDC bit along the drilling direction (g represents the gravity acceleration, 9.8 m/s^2). It can be gained that the acceleration fluctuates continuously in the whole drilling process, which shows that drill-bit always keeps on vibrating along the drilling direction. When rock is broken and rock debris is formed, the drill-bit loses support so that its axial acceleration suddenly increases. The acceleration will not decrease and even reverse until the drill-bit keeps contact with the undamaged rock again. This is the main reason for the axial acceleration fluctuation of drill-bit during the rock-breaking process.

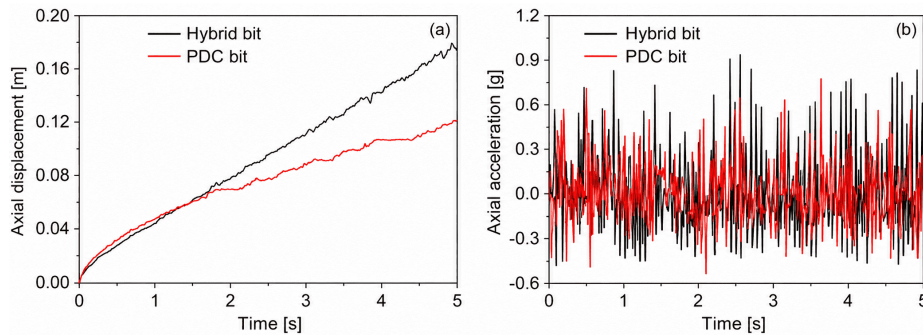


Figure 10. Axial response comparisons of hybrid bit and polycrystalline diamond compact (PDC) bit: (a) axial displacement; (b) axial acceleration

Using the data within $t = 2\text{--}5$ s in Figure 10b, the axial acceleration RMS of the hybrid bit is 0.337 g , which is 1.39 times that of the PDC bit. This indicates that the hybrid bit has a greater axial impact on rock than the PDC bit under the same drilling conditions. The main reason for this phenomenon is one or two teeth of the roller-cone structure contact with rock in turn during hybrid bit penetrating rock formation [19], causing an additional axial vibration to the hybrid bit. The consequential impact dynamic load helps to promote the emergence, expansion and penetration of cracks within rock, leading to brittle failure and thus increasing rock-breaking efficiency [20].

However, an excessive impact load can also cause the rapid damage of drilling tools. Therefore, the axial vibration levels of hybrid bit and PDC bit need to be quantified. The axial vibration grading criteria proposed by the Baker Hughes INTEQ is shown in Table 2, where the axial vibration levels are measured in the multiples of g , which is based on the RMS of the actual measured axial acceleration. Based on the axial vibration grading criteria proposed by Baker Hughes INTEQ, axial vibration levels from 0 to 2 are considered safe [21]. If the axial vibration is at level 3–4, the cumulative working time of a drilling tool should not exceed 3 h. If the axial vibration is above level 4, the cumulative working time should not exceed 20 min. According to the simulation results shown in Figure 10b, the axial vibrations of hybrid bit and PDC bit are at the 0 level, which indicates that the application of hybrid bit can increase the axial vibration amplitude without essentially leading to the rapid failure of drilling tool, thus further improving rock-breaking efficiency.

Table 2. The axial vibration grading criteria

Rank	0	1	2	3	4	5	6	7
Vibration level	0–0.5 g	0.5–1 g	1–2 g	2–3 g	3–5 g	5–8 g	8–15 g	more than 15 g

5 Field Drilling Experiment

In order to further evaluate the performance of the PDC-cone hybrid bit under real drilling conditions, an 8.5-inch hybrid bit consisting of four PDC blades and two roller-cones was designed and manufactured, as shown in Figure 11. Field testing was conducted in a 2820 m deep test well, and the drilling performance was compared with that of a previous offset well drilled using an 8.5-inch PDC bit with five blades. The test well is located adjacent to the offset well, and both wells penetrate similar lithological formations. The comparison is considered valid because the two wells were drilled with comparable drilling parameters, including WOB, rotary speed, mud properties, and bit size, allowing the ROP to be used as a reliable indicator of bit performance. Detailed drilling parameters of the test well are as follows: WOB 2–12 tons, rotary speed 170 r/min, drilling interval 2821–2860 m (39 m total footage), drilling time 10 hours, mud density 1.46 g/cm^3 , mud viscosity 48 s, and mud pH 9.5. For the offset well, the key drilling parameters were as follows: WOB 2–14 tons, rotary speed 160 r/min, drilling interval 2805–2850 m (45 m total footage), mud density 1.42 g/cm^3 , and mud viscosity 46 s.



Figure 11. Design and manufacture of 8.5-inch polycrystalline diamond compact (PDC)-cone hybrid bit

In the test well, the PDC-cone hybrid bit penetrates in rock 39 m at a ROP of 3.9 m/hr. The hybrid bit is still in relatively excellent condition after drilling operation, without severely worn and broken cutters, as shown in Figure 12. This again confirms that the application of hybrid bit can effectively reduce torsional vibration and stick-slip effects, alleviating the failure of drilling tools and prolonging their service life.



Figure 12. Polycrystalline diamond compact (PDC)-cone hybrid bit after drilling the test well

6 Conclusions

This work has presented a detailed numerical simulation and field experiment of rock-breaking PDC-cone hybrid bit. Firstly, the rock-breaking mechanism of hybrid bit is analyzed. On this basis, the 3D FE modeling of hybrid bit and PDC bit penetration rock formation is performed. In this process, the 3D uniaxial compression experiment and simulation are carried out to calibrate and determine the key property parameters of the rock. The torsional and axial dynamics characteristics of hybrid bit and PDC bit are then quantified for comparison. In general, three main conclusions can be drawn as follows:

(1) During the rock-breaking process of the hybrid bit, the roller-cone structure generates primary crushing in the rock, and the PDC cutting structure then continuously scrapes the already damaged rock. These two actions are complementary and collectively enhance the drilling efficiency. Consequently, the hybrid bit exhibits better drilling performance than the PDC bit under the tested conditions. Its advantage over conventional roller-cone bits is also indicated by the improved overall efficiency, although the present study focuses primarily on comparison with the PDC bit.

(2) Under the same drilling conditions, the torque fluctuation of hybrid bit is as much as 50% lower than that of PDC bit, which is qualitatively consistent with the field observations. Meanwhile, the torsional vibration of PDC bit is more intense than that of hybrid, and the typical stick-slip phenomenon occurs with PDC bits, but not with hybrid bit. So, the application of PDC-cone hybrid bit helps to mitigate torsional vibration and consequently mitigate stick-slip phenomenon.

(3) Because one or two cutters of the roller-cone structure contact with the bottom-hole rock in turn during drilling process, hybrid bit has a greater axial impact on rock than PDC bit under the same drilling conditions, which helps to promote the emergence, expansion and penetration of cracks within rock, thus increasing rock-breaking efficiency. In the whole drilling process, both hybrid bit and PDC bit are in a safe environment, so the application of hybrid bit can increase the axial vibration amplitude without essentially leading to the rapid failure of drilling tool, thus further improving rock-breaking efficiency.

Author Contributions

Conceptualization, X.D.; methodology, H.C. and X.D.; writing—original draft preparation, H.C.; writing—review and editing, X.D.; visualization, H.C. All authors have read and agreed to the published version of the manuscript.

Data Availability

The data used to support the research findings are available from the corresponding author upon request.

Conflicts of Interest

The authors declare no conflicts of interest.

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